

Employee Management System (EMS)

Full Project Documentation & Source Code

Project Goal: Transition from scattered data to a centralized platform.

Key Metrics: Retrieval time < 10s, HR efficiency improved by 50%.

1. Database Layer (MySQL)

Run these scripts in your MySQL Workbench to set up the indexed schema.

```
CREATE DATABASE EmployeeDB;
USE EmployeeDB;

CREATE TABLE departments (
    dept_id INT PRIMARY KEY AUTO_INCREMENT,
    dept_name VARCHAR(100) NOT NULL
);

CREATE TABLE employees (
    emp_id INT PRIMARY KEY AUTO_INCREMENT,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100) UNIQUE,
    dept_id INT,
    hire_date DATE,
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id),
    INDEX (first_name, last_name)
);

CREATE TABLE attendance (
    att_id INT PRIMARY KEY AUTO_INCREMENT,
    emp_id INT,
    work_date DATE,
    status ENUM('Present', 'Absent', 'Leave'),
    FOREIGN KEY (emp_id) REFERENCES employees(emp_id)
);
```

```
INSERT INTO departments (dept_name) VALUES ('HR'), ('Engineering'), ('Sales');
```

2. Java Backend Source Code

DBConnection.java

```
import java.sql.Connection;
import java.sql.DriverManager;

public class DBConnection {
    private static final String URL = "jdbc:mysql://localhost:3306/EmployeeDB";
    private static final String USER = "root";
    private static final String PASS = "yourpassword"; // Update this!

    public static Connection getConnection() throws Exception {
        return DriverManager.getConnection(URL, USER, PASS);
    }
}
```

EmployeeDAO.java

```
import java.sql.*;
import java.io.*;

public class EmployeeDAO {
    public void addEmployee(String fname, String lname, String email, int deptId) {
        String query = "INSERT INTO employees (first_name, last_name, email, dept_id, hire_date) VALUES (?, ?, ?, ?, ?)";
        try (Connection conn = DBConnection.getConnection();
            PreparedStatement pstmt = conn.prepareStatement(query)) {
            pstmt.setString(1, fname);
            pstmt.setString(2, lname);
            pstmt.setString(3, email);
            pstmt.setInt(4, deptId);
            pstmt.executeUpdate();
            System.out.println("Employee record digitized successfully.");
        } catch (Exception e) { e.printStackTrace(); }
    }

    public void searchEmployee(String name) {
```

```

        String query = "SELECT * FROM employees WHERE first_name LIKE ? OR last_name LIKE ?";
        try (Connection conn = DBConnection.getConnection();
            PreparedStatement pstmt = conn.prepareStatement(query)) {
            pstmt.setString(1, "%" + name + "%");
            pstmt.setString(2, "%" + name + "%");
            ResultSet rs = pstmt.executeQuery();
            while (rs.next()) {
                System.out.println(rs.getInt("emp_id") + " | " + rs.getString("first_name"));
            }
        } catch (Exception e) { e.printStackTrace(); }
    }

    public void generateAttendanceReport(int empId) {
        String query = "SELECT COUNT(*) as total_present FROM attendance WHERE emp_id = ?";
        try (Connection conn = DBConnection.getConnection();
            PreparedStatement pstmt = conn.prepareStatement(query)) {
            pstmt.setInt(1, empId);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                String report = "Employee ID " + empId + " presence count: " + rs.getInt("total_present");
                System.out.println(report);
                exportReportToFile(report);
            }
        } catch (Exception e) { e.printStackTrace(); }
    }

    private void exportReportToFile(String content) {
        try (PrintWriter out = new PrintWriter(new FileWriter("HR_Report.txt", true))) {
            out.println(content);
            System.out.println("Report exported to HR_Report.txt");
        } catch (IOException e) { e.printStackTrace(); }
    }
}

```

Main.java

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        EmployeeDAO dao = new EmployeeDAO();
        Scanner sc = new Scanner(System.in);
    }
}

```

```

while (true) {
    System.out.println("\n--- Employee Management System ---");
    System.out.println("1. Add Employee");
    System.out.println("2. Search Employee");
    System.out.println("3. Generate Attendance Report & Export");
    System.out.println("4. Exit");
    System.out.print("Select choice: ");

    int choice = sc.nextInt();
    sc.nextLine();

    if (choice == 4) break;

    switch (choice) {
        case 1:
            System.out.print("First Name: "); String fn = sc.nextLine();
            System.out.print("Last Name: "); String ln = sc.nextLine();
            System.out.print("Email: "); String email = sc.nextLine();
            System.out.print("Dept ID: "); int dId = sc.nextInt();
            dao.addEmployee(fn, ln, email, dId);
            break;
        case 2:
            System.out.print("Search Name: "); String name = sc.nextLine();
            dao.searchEmployee(name);
            break;
        case 3:
            System.out.print("Employee ID: "); int id = sc.nextInt();
            dao.generateAttendanceReport(id);
            break;
    }
}
}
}

```